Binary Tree Solutions

Solution 1:

Time Complexity : o(h)
Space Complexity: o(1)

```
import java.util.*;
class Solution{

static class Node{
        int data;
        Node left;
        Node right;
};

static Node newNode(int data){
        Node temp = new Node();
        temp.data = data;
        temp.left = temp.right = null;
        return (temp);
}

static boolean isUnivalTree(Node root){

        if (root == null){
                return true;
        }

        if (root.left != null
                && root.data != root.left.data)
                return false;

        if (root.right != null
                && root.data != root.right.data)
                return false;

        return isUnivalTree(root.left)
                && isUnivalTree(root.right);
```

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```
}

public static void main(String[] args){

        Node root = newNode(1);
        root.left = newNode(1);
        root.right = newNode(1);
        root.left.left = newNode(1);
        root.left.right = newNode(1);
        root.right.right = newNode(1);

        if (isUnivalTree(root)) {
                System.out.print("YES");
        }
        else{
                System.out.print("NO");
        }
}
}
```

Solution 2 :

Time Complexity : o(n)
Space Complexity: o(n)

```
class Node{
        int data;
        Node left, right;

        public Node(int item)
        {
                data = item;
                left = right = null;
        }
}

class Solution{
        Node root;

        void mirror(){
                root = mirror(root);
        }
```

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```
Node mirror(Node node){
        if (node == null)
                return node;

        /* do the subtrees */
        Node left = mirror(node.left);
        Node right = mirror(node.right);

        /* swap the left and right pointers */
        node.left = right;
        node.right = left;

        return node;
}

void inOrder(){
        inOrder(root);
}

void inOrder(Node node){
        if (node == null)
                return;

        inOrder(node.left);
        System.out.print(node.data + " ");

        inOrder(node.right);
}

public static void main(String args[]){
        BinaryTree tree = new BinaryTree();
        tree.root = new Node(1);
        tree.root.left = new Node(2);
        tree.root.right = new Node(3);
        tree.root.left.left = new Node(4);
        tree.root.left.right = new Node(5);

        System.out.println("Inorder traversal of input tree is :");
        tree.inOrder();
        System.out.println("");
        tree.mirror();
        System.out.println("Inorder traversal of binary tree is : ");
        tree.inOrder();

}
```

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```
}
```

Solution 3 :

Time Complexity : o(n)
Space Complexity: o(1)

```
class Solution {

static class Node {
        int data;
        Node left, right;
}

static Node newNode(int data){
        Node newNode = new Node();
        newNode.data = data;
        newNode.left = null;
        newNode.right = null;
        return (newNode);
}

static Node deleteLeaves(Node root, int x){
        if (root == null)
                return null;
        root.left = deleteLeaves(root.left, x);
        root.right = deleteLeaves(root.right, x);

        if (root.data == x && root.left == null && root.right == null) {

                return null;
        }
        return root;
}

static void inorder(Node root){
        if (root == null)
                return;
        inorder(root.left);
```

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```
        System.out.print(root.data + " ");
        inorder(root.right);
}

public static void main(String[] args){
        Node root = newNode(10);
        root.left = newNode(3);
        root.right = newNode(10);
        root.left.left = newNode(3);
        root.left.right = newNode(1);
        root.right.right = newNode(3);
        root.right.right.left = newNode(3);
        root.right.right.right = newNode(3);
        deleteLeaves(root, 3);
        System.out.print("Inorder traversal after deletion : ");
        inorder(root);
}
}
```

Solution 4 :

Time Complexity : o(n*n)
Space Complexity: o(n*n)

```
import java.util.HashMap;
public class Solution {

        static HashMap<String, Integer> m;
        static class Node {
                int data;
                Node left;
                Node right;
                Node(int data){
                        this.data = data;
                        left = null;
                        right = null;
                }
        }
```

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```
    static String inorder(Node node){
        if (node == null)
            return "";

        String str = "(";
        str += inorder(node.left);
        str += Integer.toString(node.data);
        str += inorder(node.right);
        str += ")";

        if (m.get(str) != null && m.get(str)==1 )
            System.out.print( node.data + " ");

        if (m.containsKey(str))
            m.put(str, m.get(str) + 1);
        else
            m.put(str, 1);

        return str;
    }

    static void printAllDups(Node root){
        m = new HashMap<>();
        inorder(root);
    }

    public static void main(String args[]){
        Node root = null;
        root = new Node(1);
        root.left = new Node(2);
        root.right = new Node(3);
        root.left.left = new Node(4);
        root.right.left = new Node(2);
        root.right.left.left = new Node(4);
        root.right.right = new Node(4);
        printAllDups(root);
    }
}
```

Solution 5 :

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Time Complexity : o(n)
Space Complexity: o(1)

```
class Node {

	int data;
	Node left, right;

	public Node(int item) {
		data = item;
		left = right = null;
	}
}

class Res {
	public int val;
}

class Solution {
	Node root;
	int findMaxUtil(Node node, Res res)  {

		if (node == null)
			return 0;

		int l = findMaxUtil(node.left, res);
		int r = findMaxUtil(node.right, res);

		int max_single = Math.max(Math.max(l, r) + node.data,
		                                                node.data);

		int max_top = Math.max(max_single, l + r + node.data);

		res.val = Math.max(res.val, max_top);

		return max_single;
	}

	int findMaxSum() {
```

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```
        return findMaxSum(root);
    }

    int findMaxSum(Node node) {

        Res res = new Res();
        res.val = Integer.MIN_VALUE;

        findMaxUtil(node, res);
        return res.val;
    }

    public static void main(String args[]) {
        Solution tree = new Solution();
        tree.root = new Node(10);
        tree.root.left = new Node(2);
        tree.root.right = new Node(10);
        tree.root.left.left = new Node(20);
        tree.root.left.right = new Node(1);
        tree.root.right.right = new Node(-25);
        tree.root.right.right.left = new Node(3);
        tree.root.right.right.right = new Node(4);
        System.out.println("maximum path sum is : " +
                                                tree.findMaxSum());
    }
}
```

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